

Exercise 9 - MCMC - Gibbs Sampling, Metropolis-Hastings Sampling

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Setup

```
set.seed(12242987)
library(coda)

## Warning: package 'coda' was built under R version 4.4.2

rho <- 0.5
M <- 30000
```

1. Implement a Gibbs sampler to sample from this distribution.

In this implementation i used the init parameter to pass tthe initial miu values. The variance resulting in the rnorm function is given by the covariance matrix.

```
gibbs_sampler <- function(rho, M, init = c(0, 0)) {
  theta <- matrix(0, nrow = M, ncol = 2)
  theta[1, ] <- init

  for (i in 2:M) {
    theta2 <- theta[i - 1, 2]
    mean1 <- rho * theta2
    theta[i, 1] <- rnorm(1, mean = mean1, sd = sqrt(1 - rho^2))

    theta1 <- theta[i, 1]
    mean2 <- rho * theta1
    theta[i, 2] <- rnorm(1, mean = mean2, sd = sqrt(1 - rho^2))
  }
  return(theta)
}
```

2. Use the Metropolis-Hastings algorithm with block-wise update to simulate from this distribution.

In this implementation the pdf was implemented in the acceptance criteria while we use the same default miu values in the init parameter.

```
metropolis_hastings <- function(rho, M, init = c(0, 0)) {
  theta <- matrix(0, nrow = M, ncol = 2)
```

```

theta[1, ] <- init

for (i in 2:M) {
  theta_proposed <- rnorm(2, mean = theta[i - 1, ], sd = 1)

  num <- exp(-1 / (2 * (1 - rho^2)) *
    (theta_proposed[1]^2 + theta_proposed[2]^2 -
     2 * rho * theta_proposed[1] * theta_proposed[2]))
  den <- exp(-1 / (2 * (1 - rho^2)) *
    (theta[i - 1, 1]^2 + theta[i - 1, 2]^2 -
     2 * rho * theta[i - 1, 1] * theta[i - 1, 2]))
  acceptance_ratio <- num / den

  if (runif(1) < acceptance_ratio) {
    theta[i, ] <- theta_proposed
  } else {
    theta[i, ] <- theta[i - 1, ]
  }
}
return(theta)
}

```

3. Perform chain diagnostics on the resulting chain.

For chain diagnostics i implemented trace plots and density plots

```

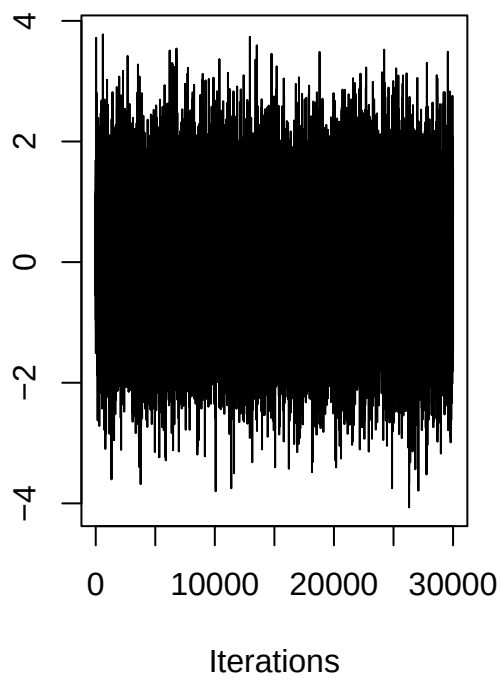
gibbs_chain <- gibbs_sampler(rho, M)

mh_chain <- metropolis_hastings(rho, M)
gibbs_mcmc <- as.mcmc(gibbs_chain)
mh_mcmc <- as.mcmc(mh_chain)

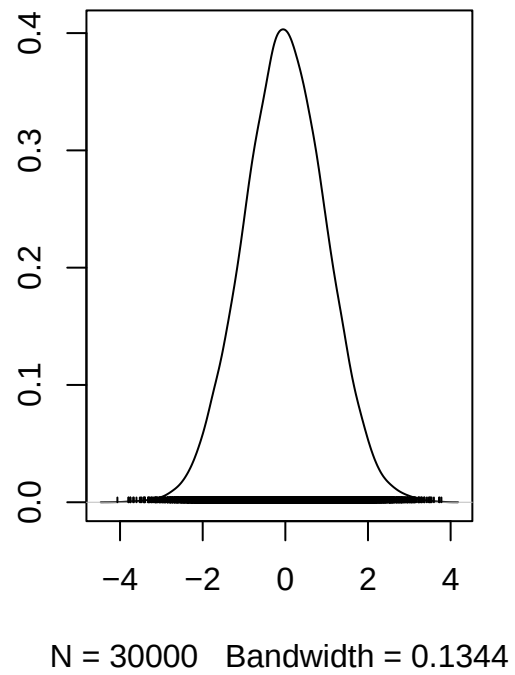
# Plot trace and density plots
plot(gibbs_mcmc[, 1], main = "Gibbs: Trace Plot for Theta1", type = "l")

```

Gibbs: Trace Plot for Theta1

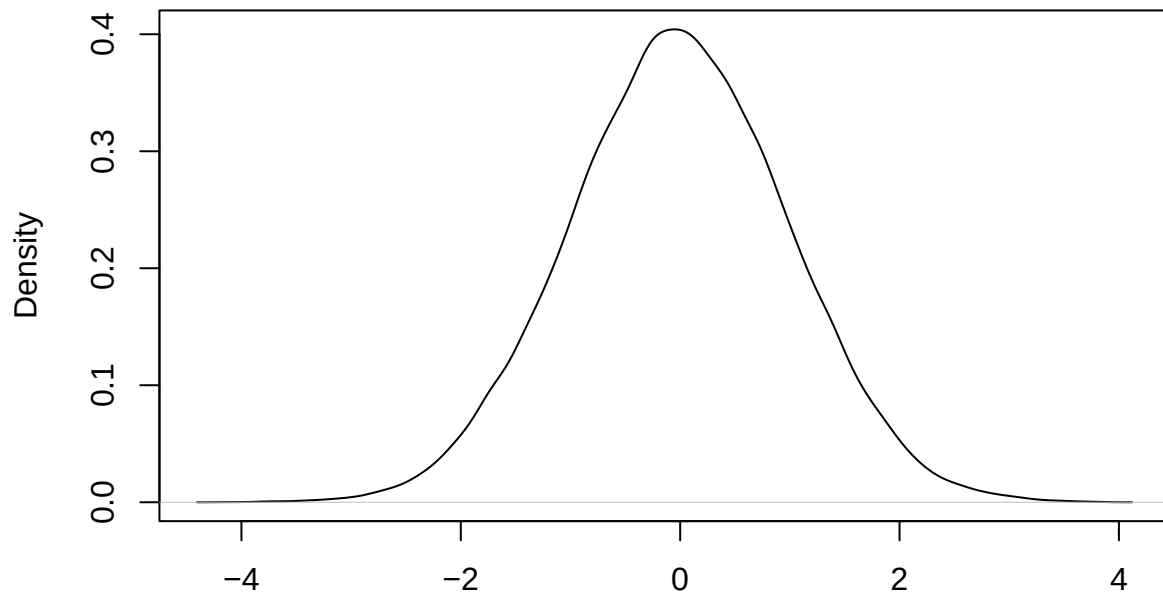


Gibbs: Trace Plot for Theta1



```
plot(density(gibbs_chain[, 1]), main = "Gibbs: Density Plot for Theta1")
```

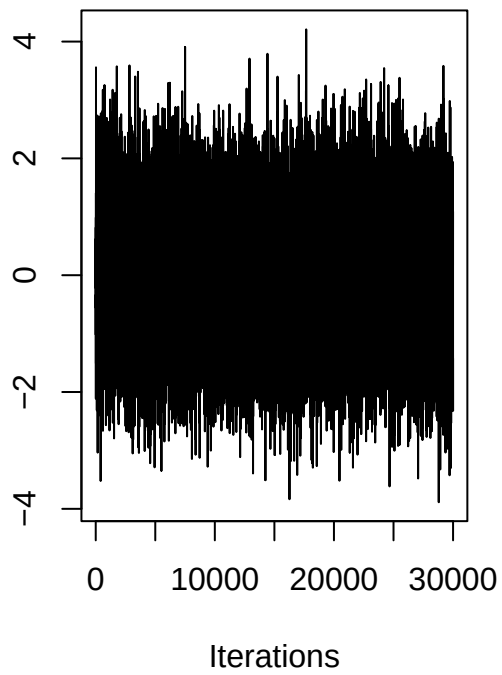
Gibbs: Density Plot for Theta1



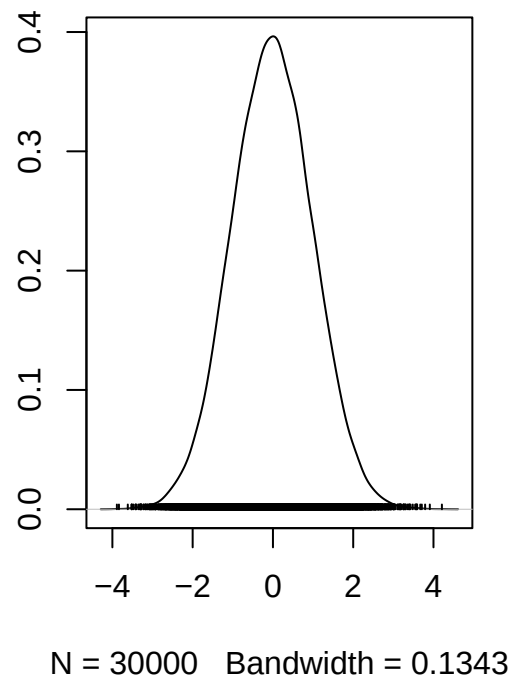
N = 30000 Bandwidth = 0.1141

```
plot(gibbs_mcmc[, 2], main = "Gibbs: Trace Plot for Theta2", type = "l")
```

Gibbs: Trace Plot for Theta2

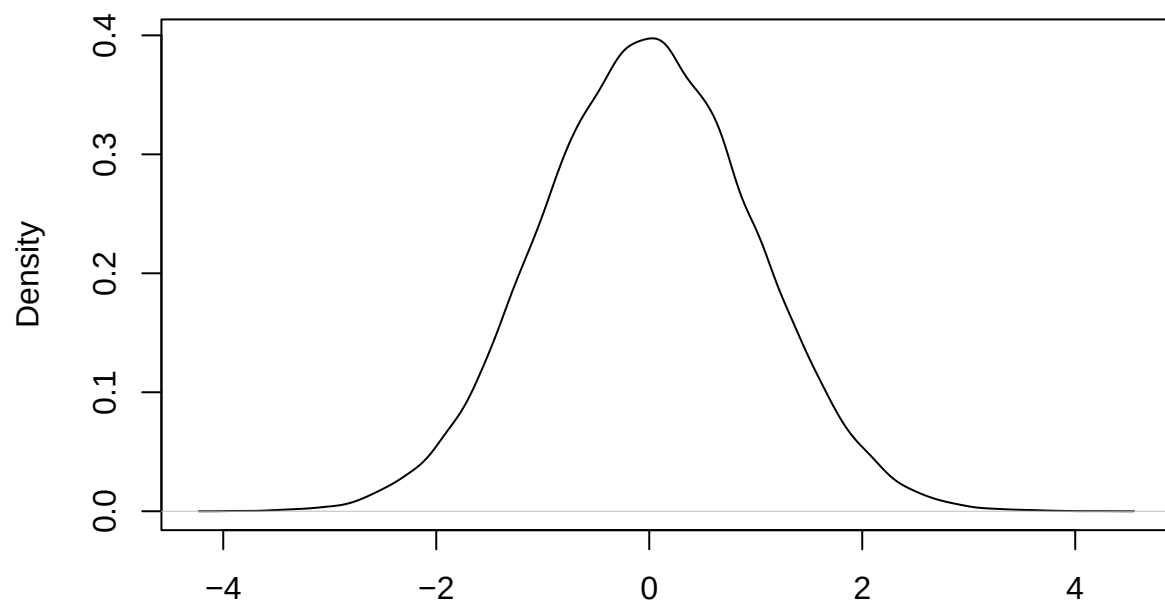


Gibbs: Trace Plot for Theta2



```
plot(density(gibbs_chain[, 2]), main = "Gibbs: Density Plot for Theta2")
```

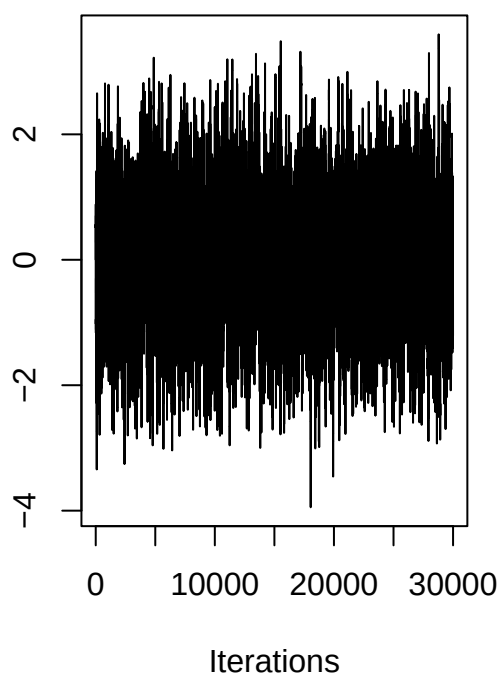
Gibbs: Density Plot for Theta2



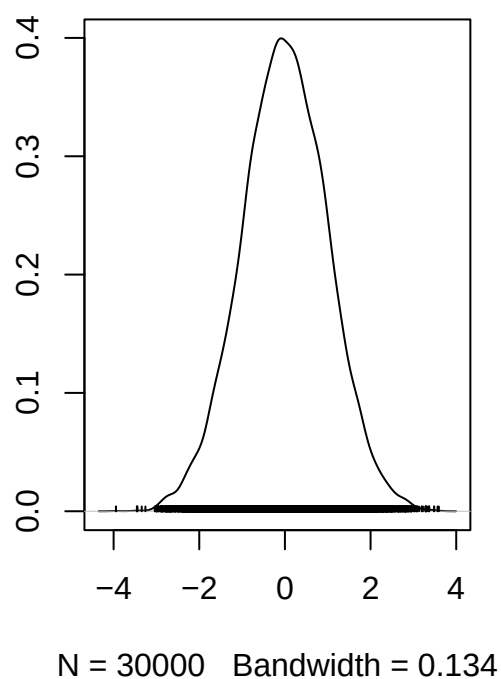
N = 30000 Bandwidth = 0.114

```
# Repeat for Metropolis-Hastings  
plot(mh_mcmc[, 1], main = "MH: Trace Plot for Theta1", type = "l")
```

MH: Trace Plot for Theta1

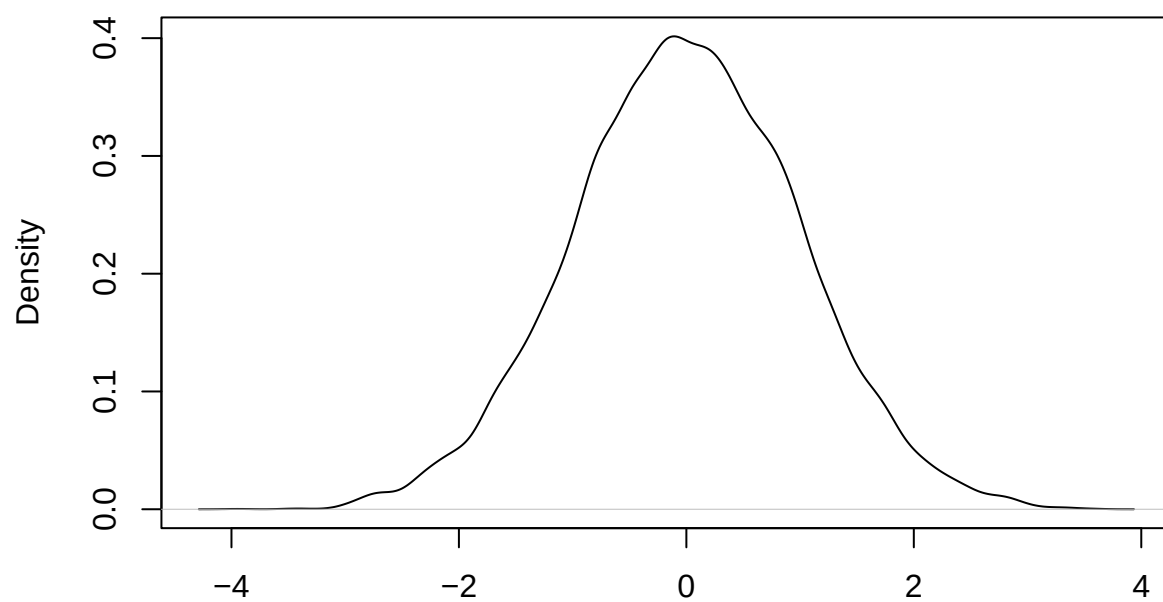


MH: Trace Plot for Theta1



```
plot(density(mh_chain[, 1]), main = "MH: Density Plot for Theta1")
```

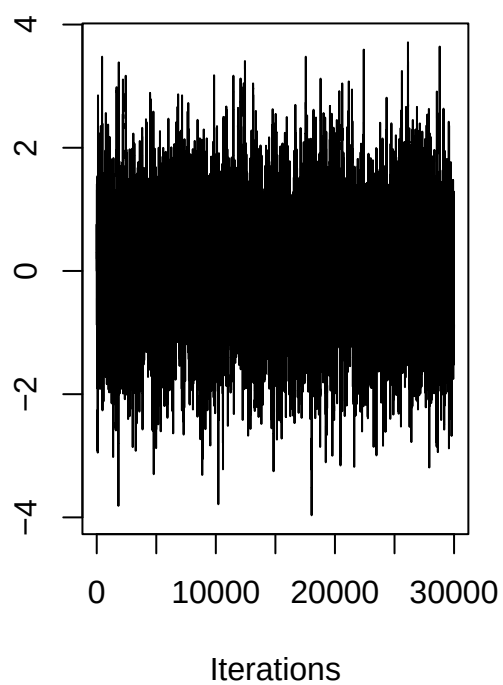
MH: Density Plot for Theta1



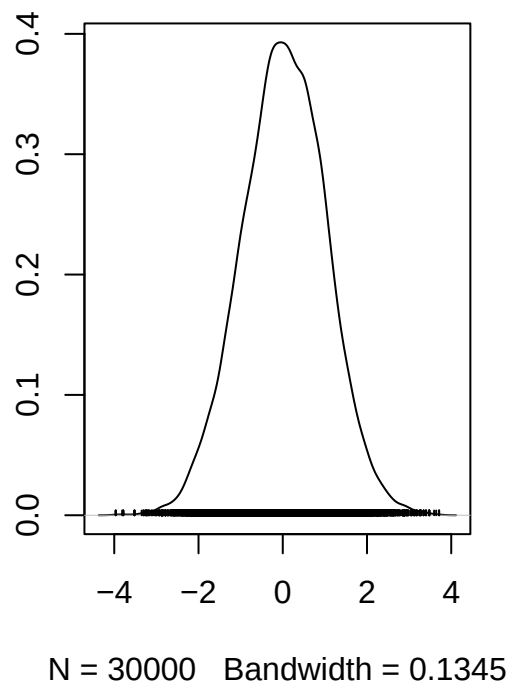
N = 30000 Bandwidth = 0.1138

```
plot(mh_mcmc[, 2], main = "MH: Trace Plot for Theta2", type = "l")
```

MH: Trace Plot for Theta2

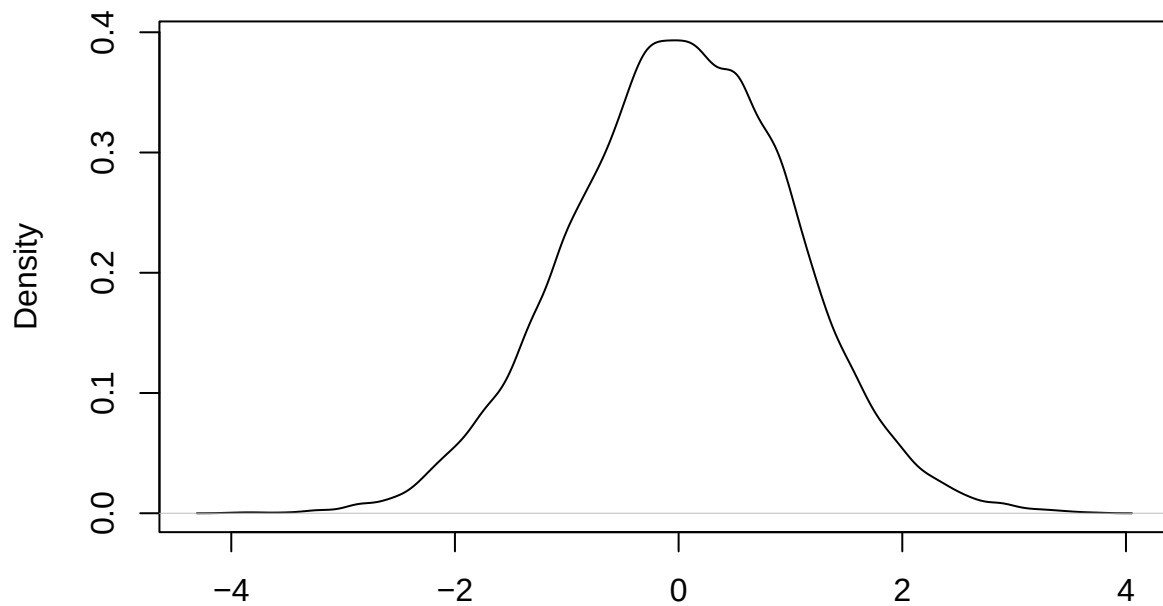


MH: Trace Plot for Theta2



```
plot(density(mh_chain[, 2]), main = "MH: Density Plot for Theta2")
```

MH: Density Plot for Theta2



N = 30000 Bandwidth = 0.1142

We can say that the trace plots show that the distributions explore the parameter space properly because they converge towards the middle value zero while keeping their randomness when generated by looking very jagged. The density plots are smooth and show the shape of the distribution which is close to the normal distribution.